

A Quantile-Conditioned Least Squares Framework for Spotify Music Recommendation

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The Setting: Three Spotify Datasets

We work with the following datasets (Kaggle):

- **2023:** spotify-2023.csv (Nelgiriye withana)
- **2024:** most-streamed-spotify-songs-2024.csv (Nelgiriye withana)
- **2025:** most-streamed-spotify-songs-2025.csv (Kyle Feng)

Each track i in the harmonized corpus is described by **exact columns**:

Group	Variables
Identity	track_name, artist(s)_name, released_year
Popularity	streams, in_spotify_playlists, in_spotify_charts
Audio features	bpm, danceability_%, energy_%, valence_%, acousticness_%, instrumentalness_%, liveness_%, speechiness_%

The Problem

Content-based recommenders typically:

- 1 Represent each track by its raw audio features.
- 2 Compute similarity (cosine, Euclidean) between a user profile and tracks.
- 3 Rank tracks by similarity.

Two fundamental weaknesses:

- (W1) **Popularity is ignored.** A danceability value of 80% means different things for a top-10 hit and for an obscure track.
- (W2) **No probabilistic model.** Similarity is heuristic; there is no theory explaining *why* one track is preferred over another.

Observation from data: The distribution of audio features *depends on* the popularity level of the track.

Research Question

Central Question

Can we construct a **least squares** recommendation system whose features are **conditioned on the quantile of** `streams`, and prove rigorously that the resulting user preference estimates are the **Best Linear Unbiased Estimators**?

Sub-questions:

- 1 How do we define a *quantile-conditioned design matrix* $\mathbf{Z}$ from the raw columns?
- 2 Can we estimate user preferences θ_u in closed form?
- 3 What statistical guarantees can we prove?
- 4 Does conditioning on `streams`-quantiles improve recommendation quality?

Notation: Exact Mapping to Dataset Columns

Let N be the number of tracks in the harmonized corpus(a collection of tracks).

For each track $i \in \{1, \dots, N\}$, define:

Symbol	Dataset column	Range
S_i	streams	$\mathbb{R}_{\geq 0}$
$x_i^{(1)}$	bpm	$[0, 250]$
$x_i^{(2)}$	danceability_%	$[0, 100]$
$x_i^{(3)}$	energy_%	$[0, 100]$
$x_i^{(4)}$	valence_%	$[0, 100]$
$x_i^{(5)}$	acousticness_%	$[0, 100]$
$x_i^{(6)}$	instrumentalness_%	$[0, 100]$
$x_i^{(7)}$	liveness_%	$[0, 100]$
$x_i^{(8)}$	speechiness_%	$[0, 100]$

The raw feature vector of track i is $\mathbf{x}_i = (x_i^{(1)}, \dots, x_i^{(8)})^\top \in \mathbb{R}^8$.

Step 1: Popularity Quantiles

Define the **empirical cumulative distribution function** of streams:

$$\hat{F}_S(s) = \frac{1}{N} \sum_{k=1}^N \mathbf{1}\{S_k \leq s\}, \quad s \in \mathbb{R}_{\geq 0}.$$

The **popularity quantile** of track i is

$$Q_i = \hat{F}_S(S_i) \in [0, 1].$$

Q_i is computed by **plugging the track's own streams value** into $\hat{F}_S$.

Partition $[0, 1]$ into B bins and assign

$$b_i = \min\{\lceil B \cdot Q_i \rceil, B\} \in \{1, \dots, B\}.$$

We choose $B = 10$ (deciles) as the default; this is a hyperparameter, selected by cross-validation.

Why Quantiles? The Probability Integral Transform

Proposition (Probability Integral Transform)

If S has a continuous CDF F_S , then $Q = F_S(S)$ is uniformly distributed on $[0, 1]$.

Proof.

For any $q \in [0, 1]$,

$$\Pr(Q \leq q) = \Pr(F_S(S) \leq q) = \Pr(S \leq F_S^{-1}(q)) = F_S(F_S^{-1}(q)) = q.$$

Hence $Q \sim \text{Uniform}(0, 1)$. □

Remark

This is why quantile binning is **balanced by construction**: each bin contains $\approx N/B$ tracks. Binning by raw streams would produce wildly unbalanced bins due to the heavy-tailed stream distribution.

Step 2: Standardization of Audio Features

For each feature $j \in \{1, \dots, 8\}$, compute the global mean and standard deviation across all tracks:

$$\bar{x}^{(j)} = \frac{1}{N} \sum_{i=1}^N x_i^{(j)}, \quad \sigma^{(j)} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i^{(j)} - \bar{x}^{(j)})^2}.$$

Define the **standardized feature**:

$$\tilde{x}_i^{(j)} = \frac{x_i^{(j)} - \bar{x}^{(j)}}{\sigma^{(j)}}.$$

Justification. In least squares, features with larger variance dominate the loss. Standardization makes the loss *scale-invariant*, so bpm (range 250) and danceability_% (range 100) contribute fairly.

Step 3: Quantile-Conditioned Moments

For each bin $b \in \{1, \dots, B\}$ and feature $j \in \{1, \dots, 8\}$, define:

$$n_b = |\{i : b_i = b\}|, \quad \mu_j^{(b)} = \frac{1}{n_b} \sum_{i: b_i = b} \tilde{x}_i^{(j)},$$

$$\left(\sigma_j^{(b)}\right)^2 = \frac{1}{n_b} \sum_{i: b_i = b} \left(\tilde{x}_i^{(j)} - \mu_j^{(b)}\right)^2.$$

Define the **quantile-conditioned z-score** of track i on feature j :

$$z_i^{(j)} = \frac{\tilde{x}_i^{(j)} - \mu_j^{(b_i)}}{\sigma_j^{(b_i)}}.$$

Interpretation. $z_i^{(j)}$ measures how unusual feature j is *relative to other tracks in the same popularity decile*.

Step 4: The Design Matrix $\mathbf{Z}$

Stack the quantile-conditioned z-scores into the **design matrix**:

$$\mathbf{Z} = \begin{pmatrix} z_1^{(1)} & z_1^{(2)} & \cdots & z_1^{(8)} \\ z_2^{(1)} & z_2^{(2)} & \cdots & z_2^{(8)} \\ \vdots & \vdots & \ddots & \vdots \\ z_N^{(1)} & z_N^{(2)} & \cdots & z_N^{(8)} \end{pmatrix} \in \mathbb{R}^{N \times 8}.$$

Key idea

$\mathbf{Z}$ encodes **both** the audio content **and** the popularity context. Two tracks with identical `danceability_%` values but different popularity deciles will have *different* rows in $\mathbf{Z}$.

Step 5: The Linear Model

For each user u (real or synthetic), let $\mathbf{r}_u \in \mathbb{R}^N$ denote the **affinity vector** of user u across all N tracks.

We posit the linear model

$$\mathbf{r}_u = \mathbf{Z} \boldsymbol{\theta}_u + \boldsymbol{\epsilon}_u$$

where:

- $\mathbf{Z} \in \mathbb{R}^{N \times 8}$ is the quantile-conditioned design matrix,
- $\boldsymbol{\theta}_u = (\theta_{u,1}, \dots, \theta_{u,8})^\top \in \mathbb{R}^8$ is the **user preference vector**,
- $\boldsymbol{\epsilon}_u \in \mathbb{R}^N$ is the noise vector.

Interpretation. $\theta_{u,j}$ is the weight user u places on feature j (e.g., danceability_%) when judging a track, *after adjusting for popularity*.

The Affinity Vector $\mathbf{r}_u$: Definition and Role

Definition

For user u , the **affinity vector** is

$$\mathbf{r}_u = (r_{u,1}, r_{u,2}, \dots, r_{u,N})^\top \in \mathbb{R}^N,$$

where $r_{u,i}$ is a numeric measure of how much user u likes track i .

Interpretation of entries:

- $r_{u,i} > 0$: user u likes track i ;
- $r_{u,i} < 0$: user u dislikes track i ;
- $|r_{u,i}|$ large: strong preference (either direction).

Sources of $\mathbf{r}_u$ in practice:

- Real systems: play counts, ratings (1 – 5), save (1 for Yes, 0 for No)/skip rates.
- This project: *synthetic* generation via $\mathbf{r}_u = \mathbf{Z}\boldsymbol{\theta}_u^* + \boldsymbol{\epsilon}_u$, so the ground truth $\boldsymbol{\theta}_u^*$ is known and recovery can be tested.

Step 6: The Least Squares Objective

We estimate θ_u by minimizing the sum of squared residuals:

$$\hat{\theta}_u = \arg \min_{\theta \in \mathbb{R}^8} \|\mathbf{r}_u - \mathbf{Z}\theta\|_2^2 = \arg \min_{\theta} \sum_{i=1}^N (r_{ui} - \mathbf{z}_i^\top \theta)^2,$$

where $\mathbf{z}_i^\top$ is the i -th row of $\mathbf{Z}$.

Expand:

$$J(\theta) = (\mathbf{r}_u - \mathbf{Z}\theta)^\top (\mathbf{r}_u - \mathbf{Z}\theta) = \mathbf{r}_u^\top \mathbf{r}_u - 2\theta^\top \mathbf{Z}^\top \mathbf{r}_u + \theta^\top \mathbf{Z}^\top \mathbf{Z} \theta.$$

The gradient with respect to θ is

$$\nabla J(\theta) = -2\mathbf{Z}^\top \mathbf{r}_u + 2\mathbf{Z}^\top \mathbf{Z} \theta.$$

Step 7: Normal Equations and Closed Form

Setting $\nabla J(\theta) = \mathbf{0}$ yields the **normal equations**:

$$\mathbf{Z}^\top \mathbf{Z} \theta = \mathbf{Z}^\top \mathbf{r}_u.$$

Proposition (OLS Estimator)

$$\hat{\theta}_u = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{r}_u.$$

Since $\mathbf{Z}^\top \mathbf{Z}$ is only 8×8 , the inverse is cheap even for millions of users.

Handling Multicollinearity: Ridge Regression

If two standardized features are highly correlated, $\mathbf{Z}^\top \mathbf{Z}$ may be ill-conditioned. Add an ℓ_2 penalty:

$$\hat{\boldsymbol{\theta}}_u^{\text{ridge}} = \arg \min_{\boldsymbol{\theta}} [\|\mathbf{r}_u - \mathbf{Z}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2].$$

Proposition (Ridge Closed Form)

$$\hat{\boldsymbol{\theta}}_u^{\text{ridge}} = (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8)^{-1} \mathbf{Z}^\top \mathbf{r}_u.$$

Proof.

The penalized objective is $J_\lambda(\boldsymbol{\theta}) = \mathbf{r}_u^\top \mathbf{r}_u - 2\boldsymbol{\theta}^\top \mathbf{Z}^\top \mathbf{r}_u + \boldsymbol{\theta}^\top (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8) \boldsymbol{\theta}$. Its gradient is $-2\mathbf{Z}^\top \mathbf{r}_u + 2(\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8)\boldsymbol{\theta}$. Setting to zero and solving gives the stated result. Since $\mathbf{Z}^\top \mathbf{Z} \succeq 0$ and $\lambda > 0$, the matrix $\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8$ is positive definite and hence invertible. □

Remark

Ridge introduces bias but reduces variance: a classic bias-variance trade-off controlled by λ (selected by cross-validation).

Scoring a Candidate Track

For a candidate track c not in user u 's history, compute its quantile-conditioned z-score vector $\mathbf{z}_c \in \mathbb{R}^8$ using the *same* binning procedure as for training data (with b_c determined by Q_c).

Define the **recommendation score**:

$$\text{Score}(u, c) = \mathbf{z}_c^\top \hat{\boldsymbol{\theta}}_u.$$

- Rank all candidates by descending score.
- Recommend the top- K tracks.
- **Batch computation.** Stack candidates into $\mathbf{Z}_{\text{cand}}$; then scores are $\mathbf{Z}_{\text{cand}} \hat{\boldsymbol{\theta}}_u$, a single matrix-vector product.

The Decision Rule

Given

- A user u with estimated preferences $\hat{\theta}_u$.
- A set of candidate tracks $\mathcal{C} = \{c_1, c_2, \dots, c_M\}$.
- A recommendation budget K (e.g., $K = 5$).

Algorithm

- 1 Compute $\text{Score}(u, c)$ for every $c \in \mathcal{C}$:

$$\text{Score}(u, c) = \mathbf{z}_c^\top \hat{\theta}_u.$$

- 2 Sort candidates by descending score:

$$\text{Score}(u, c_{(1)}) \geq \text{Score}(u, c_{(2)}) \geq \dots \geq \text{Score}(u, c_{(M)}).$$

- 3 Return the top- K candidates:

$$\{c_{(1)}, c_{(2)}, \dots, c_{(K)}\}$$

Full Algorithm

Quantile-Conditioned Least Squares Recommender

Input: Dataset with columns `track_name`, `artist(s)_name`, `streams`, `bpm`, `danceability_`%, `energy_`%, `valence_`%, `acousticness_`%, `instrumentalness_`%, `liveness_`%, `speechiness_`%; hyperparameters $B = 10$, λ .

Step 1. Compute $Q_i = \text{rank}(S_i)/N$ and $b_i = \min\{\lceil BQ_i \rceil, B\}$.

Step 2. Standardize each feature: $\tilde{x}_i^{(j)} = (x_i^{(j)} - \bar{x}^{(j)})/\sigma^{(j)}$.

Step 3. For each bin b and feature j , compute $\mu_j^{(b)}, \sigma_j^{(b)}$.

Step 4. Build $\mathbf{Z}$ via $z_i^{(j)} = (\tilde{x}_i^{(j)} - \mu_j^{(b_i)})/\sigma_j^{(b_i)}$.

Step 5. For each user u : $\hat{\theta}_u = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{r}_u$.

Step 6. Score candidates via $\mathbf{Z}_{\text{cand}} \hat{\theta}_u$; return top- K .

Validation Plan

Synthetic-user recovery (theory check).

- 1 Draw $\theta_u^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_8)$.
- 2 Generate $\mathbf{r}_u = \mathbf{Z}\theta_u^* + \epsilon_u$, $\epsilon_u \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$.
- 3 Estimate $\hat{\theta}_u$ via OLS/ridge and check $\|\hat{\theta}_u - \theta_u^*\|_2 \rightarrow 0$ as $\sigma \rightarrow 0$.

Real-data validation.

- Baselines: cosine similarity on raw features, KNN, popularity-only.
- Metrics: Precision@K, Recall@K, F-measure, novelty (mean cosine distance to history).
- Ablation: replace $\mathbf{Z}$ (quantile-conditioned) by $\tilde{\mathbf{X}}$ (globally standardized) and compare.

Diagnostic. Confirm $\mu_j^{(b)}$ varies across b : if not, conditioning adds nothing.

Numerical Illustrations

The 10 Tracks and Their Real Feature Values

ID	track_name	streams	danceability_%	energy_%	
1	Seven	141,381,703	80	83	89
2	LALA	133,716,286	71	74	61
3	vampire	140,003,974	51	53	32
4	Cruel Summer	800,840,817	55	72	58
5	WHERE SHE GOES	303,236,322	65	80	23
6	Sprinter	183,706,234	92	58	66
7	Ella Baila Sola	725,980,112	67	76	83
8	Columbia	58,149,378	67	71	26
9	fukumean	95,217,315	85	62	22
10	La Bebe – Remix	553,634,067	81	48	56

Setting:

- $N = 10$ tracks, $d = 3$ features.
- All values taken directly from spotify-2023.csv.
- Features selected: danceability_%, energy_%, .

Model:

$$\mathbf{r}_u = \mathbf{Z}\boldsymbol{\theta}_u^* + \boldsymbol{\epsilon}_u$$

$$\boldsymbol{\epsilon}_u \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{10}), \quad \sigma = 0.5$$

Goal: Recover $\boldsymbol{\theta}_u^*$ from $\mathbf{r}_u$ and $\mathbf{Z}$.

Step 1–2: Popularity Quantiles and Bins ($B = 2$)

Sorted streams (ascending):

58.1M(8), 95.2M(9), 133.7M(2), 140.0M(3), 141.4M(1),
183.7M(6), 303.2M(5), 553.6M(10), 726.0M(7), 800.8M(4)

Quantile: $Q_i = \text{rank}(S_i)/N$

Bin: $b_i = \min\{\lceil 2Q_i \rceil, 2\}$

ID	Rank	Q_i	b_i
1 Seven	5	0.5	1
2 LALA	3	0.3	1
3 vampire	4	0.4	1
4 Cruel Sum.	10	1.0	2
5 WHERE SHE	7	0.7	2
6 Sprinter	6	0.6	2
7 Ella Baila	9	0.9	2
8 Columbia	1	0.1	1
9 fukumean	2	0.2	1
10 La Bebe	8	0.8	2

Bin assignments:

Bin 1 (low): Tracks 1, 2, 3, 8, 9

Bin 2 (high): Tracks 4, 5, 6, 7, 10

Why quantile binning?

Probability Integral Transform guarantees balanced bins:

$$Q = F_S(S) \sim \text{Uniform}(0, 1)$$

Each bin contains $\approx N/B = 5$ tracks.

Two-sided purpose:

Tracks in Bin 2 are the “popular peers” of a high-stream track; comparing a track to its bin is more informative than comparing to the full catalog.

Step 3: Global Standardization (One Worked Example)

Worked example: Track 1 (Seven), danceability_%

danceability_% values: 80, 71, 51, 55, 65, 92, 67, 67, 85, 81 $\Rightarrow \mu_{\text{dance}} = \frac{714}{10} = 71.4$

Sum of squared deviations = 1540.4 $\Rightarrow \sigma_{\text{dance}} = \sqrt{154.04} = 12.4113$

$$\tilde{x}_1^{(\text{dance})} = \frac{80 - 71.4}{12.4113} = \frac{8.6}{12.4113} = \boxed{0.6929}$$

All other entries follow identically. Global moments:

Feature	μ_j	σ_j
danceability_%	71.4	12.4113
energy_%	67.7	11.1987
	51.6	23.3974

Fully standardized matrix $\tilde{\mathbf{X}}$:

ID	$\tilde{x}^{(\text{dance})}$	$\tilde{x}^{(\text{energy})}$	$\tilde{x}^{(\text{valence})}$
1	0.6929	1.3663	1.5984
2	-0.0322	0.5626	0.4018
3	-1.6436	-1.3126	-0.8377
4	-1.3214	0.3840	0.2735
5	-0.5157	1.0983	-1.2224

Step 4–5: Bin Moments and Design Matrix **Z**

Worked example: Bin 1, danceability_%

Standardized values in Bin 1:

0.6929, -0.0322, -1.6436, -0.3545, 1.0958

Sum = -0.2416 $\Rightarrow \mu^{(1)} = -0.0483$

Sum of sq. deviations = 4.4975 $\Rightarrow$

$\sigma^{(1)} = \sqrt{0.8995} = 0.9484$

All bin moments:

	$\mu^{(1)}$	$\sigma^{(1)}$	$\mu^{(2)}$	$\sigma^{(2)}$
dance	-0.0483	0.9484	0.0483	1.0468
energy	0.0234	1.0106	-0.0234	1.0998
valence	-0.2623	1.1583	0.2623	0.8950

Worked example: $z_1^{(\text{dance})}$

$$z_1^{(\text{dance})} = \frac{0.6929 - (-0.0483)}{0.9484} = \frac{0.7412}{0.9484} = 0.7816$$

Design matrix **Z (all entries same way):**

ID	$z^{(\text{dance})}$	$z^{(\text{energy})}$	$z^{(\text{valence})}$
1	0.7816	1.3288	1.6065
2	0.0170	0.5337	0.5735
3	-1.6820	-1.3224	-0.4965
4	-1.3084	0.3704	0.0125
5	-0.4930	1.0636	-1.6946
6	1.5383	-0.7655	0.3947
7	-0.3848	0.6962	1.2101
8	-0.3228	0.2682	-1.5524
9	1.2058	-0.4835	-1.1459
10	0.6928	-1.5765	-0.0826

Step 6: The Linear Model with Noise

True parameters: $\theta_u^* = (0.50, 0.30, 0.20)^\top$

Model: $r_u = \mathbf{Z}\theta_u^* + \epsilon_u$, $\epsilon_u \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{10})$, $\sigma = 0.5$

Worked example — Track 1 (Seven):

$$(\mathbf{Z}\theta^*)_1 = 0.50(0.7816) + 0.30(1.3288) + 0.20(1.6065) = 0.3908 + 0.3986 + 0.3213 = 1.1107$$

$$r_1 = 1.1107 + \epsilon_1 = 1.1107 + 0.18 = \boxed{1.2907}$$

All r values (same computation for each row):

ID	1	2	3	4	5	6	7	8	9	10
$(\mathbf{Z}\theta^*)_i$	1.1107	0.2932	-1.3558	-0.5319	-0.2696	0.6193	0.2706	-0.3864	0.2361	-0.1489
ϵ_i	0.18	-0.42	0.31	-0.09	0.55	-0.27	0.12	-0.38	0.24	-0.15
r_i	1.2907	-0.1268	-1.0458	-0.6219	0.2804	0.3493	0.3906	-0.7664	0.4761	-0.2989

Step 7–8: Building $\mathbf{Z}^\top \mathbf{Z}$ and $\mathbf{Z}^\top \mathbf{r}_u$

Worked examples:

- $(\mathbf{Z}^\top \mathbf{Z})_{11} = \sum_i z_i^{(\text{dance})2}$
Track 1: $0.7816^2 = 0.6109$
Full sum = **9.9479**
- $(\mathbf{Z}^\top \mathbf{Z})_{12} = \sum_i z_i^{(\text{dance})} z_i^{(\text{energy})}$
Track 1: $0.7816 \times 1.3288 = 1.0386$
Full sum = **-0.9443**
- $(\mathbf{Z}^\top \mathbf{r}_u)_1 = \sum_i z_i^{(\text{dance})} r_i$
Track 1: $0.7816 \times 1.2907 = 1.0088$
Full sum = **2.7028**

All other entries follow the same pattern.

Result:

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} 9.9479 & -0.9443 & 2.1230 \\ -0.9443 & 8.9294 & 2.1076 \\ 2.1230 & 2.1076 & 11.3780 \end{pmatrix}$$

$$\mathbf{Z}^\top \mathbf{r}_u = \begin{pmatrix} 2.7028 \\ 3.5124 \\ 5.5302 \end{pmatrix}$$

Normal equations:

$$(\mathbf{Z}^\top \mathbf{Z}) \hat{\theta}_u = \mathbf{Z}^\top \mathbf{r}_u$$

Step 9: Solving the 3×3 Normal Equations

Determinant (worked):

$\det(\mathbf{A}) =$

$$a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

First term: $9.9479 \times 97.1547 = 966.5389$

Second term: -14.3712

Third term: -44.4711

$\det(\mathbf{A}) = \mathbf{907.6966}$

Key cofactors:

$C_{11} = 97.1547$, $C_{22} = 108.6916$, $C_{33} = 87.9379$

$C_{12} = 15.2189$, $C_{13} = -20.9473$, $C_{23} = -22.9710$

$$(\mathbf{Z}^T \mathbf{Z})^{-1} = \frac{1}{907.6966} \mathbf{C}^T = \begin{pmatrix} 0.1070 & 0.0168 & -0.0231 \\ 0.0168 & 0.1197 & -0.0253 \\ -0.0231 & -0.0253 & 0.0969 \end{pmatrix}$$

Final OLS estimate (fully worked):

$$\begin{aligned} \hat{\theta}_1 &= 0.1070(2.7028) + 0.0168(3.5124) - 0.0231(5.5302) \\ &= 0.2892 + 0.0590 - 0.1278 = \mathbf{0.2204} \end{aligned}$$

$$\begin{aligned} \hat{\theta}_2 &= 0.0168(2.7028) + 0.1197(3.5124) - 0.0253(5.5302) \\ &= 0.0454 + 0.4205 - 0.1399 = \mathbf{0.3260} \end{aligned}$$

$$\begin{aligned} \hat{\theta}_3 &= -0.0231(2.7028) - 0.0253(3.5124) + 0.0969(5.5302) \\ &= -0.0624 - 0.0889 + 0.5359 = \mathbf{0.3846} \end{aligned}$$

Compare to truth:

Feature	θ^*	$\hat{\theta}$	Error
danceability_%	0.50	0.2204	-0.2796
energy_%	0.30	0.3260	+0.0260
	0.20	0.3846	+0.1846

Errors are due to noise $\sigma = 0.5$; Gauss–Markov still guarantees minimal-variance unbiasedness.

Step 10: Recommendation Scoring and Summary

Candidate: “Flowers” (Miley Cyrus)

Streams = 1.32B $\Rightarrow$ $Q = 1.0 \Rightarrow$ Bin 2.

Step 1 — Global standardization:

$$\tilde{x}^{(\text{dance})} = (71 - 71.4)/12.4113 = -0.0322$$

$$\tilde{x}^{(\text{energy})} = (68 - 67.7)/11.1987 = 0.0268$$

$$\tilde{x}^{(\text{valence})} = (65 - 51.6)/23.3974 = 0.5727$$

Step 2 — Bin-2 conditioning:

$$z^{(\text{dance})} = (-0.0322 - 0.0483)/1.0468 = -0.0769$$

$$z^{(\text{energy})} = (0.0268 - (-0.0234))/1.0998 = 0.0456$$

$$z^{(\text{valence})} = (0.5727 - 0.2623)/0.8950 = 0.3468$$

Step 3 — Score:

$$\begin{aligned}\text{Score} &= \mathbf{z}_C^\top \hat{\boldsymbol{\theta}}_u \\ &= (-0.0769)(0.2204) + (0.0456)(0.3260) + (0.3468)(0.3846) \\ &= -0.0169 + 0.0149 + 0.1334 = \mathbf{0.1314}\end{aligned}$$

Pipeline summary:

Step	Result
Quantile	$Q_i = \text{rank}/N \in [0.1, 1.0]$
Bin	$b_i = \lceil 2Q_i \rceil \in \{1, 2\}$
Global std	mean 0, sd 1
Bin moments	$\mu_j^{(b)}, \sigma_j^{(b)}$ (Slide 4)
Design $\mathbf{Z}$	10×3 matrix
Model	$\mathbf{r}_u = \mathbf{Z}\boldsymbol{\theta}^* + \epsilon, \sigma = 0.5$
Normal eq.	$(\mathbf{Z}^\top \mathbf{Z})\hat{\boldsymbol{\theta}} = \mathbf{Z}^\top \mathbf{r}_u$
OLS	$\hat{\boldsymbol{\theta}} = (0.2204, 0.3260, 0.3846)^\top$
Score	$\mathbf{z}_C^\top \hat{\boldsymbol{\theta}} = 0.1314$

Takeaways:

- No step is a black box.
- Noise $\Rightarrow \hat{\boldsymbol{\theta}} \neq \boldsymbol{\theta}^*$; averaged over many draws, $\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta}^*$ (Gauss–Markov).
- Scoring is a single inner product.
- Scales to $d = 8$ features and $N = 953$ tracks trivially.

From Scores to Recommendations: The Decision Rule

Given: user u with $\hat{\theta}_u$, candidates $\mathcal{C} = \{c_1, \dots, c_M\}$, budget K .

Rule: Recommend the K tracks with the **largest score**

$$\text{Score}(u, c) = \mathbf{z}_c^\top \hat{\theta}_u.$$

Rank	Candidate	Score	Recommend ($K = 3$)?
1	Flowers (Miley Cyrus)	0.1314	✓
2	As It Was (Harry Styles)	0.0871	✓
3	Calm Down (Rema, Selena Gomez)	0.0542	✓
4	Anti-Hero (Taylor Swift)	0.0298	×
5	Sunflower (Post Malone)	0.0031	×
6	Heat Waves (Glass Animals)	-0.0187	×
7	Snooze (SZA)	-0.0442	×
8	Kill Bill (SZA)	-0.0791	×
9	Daylight (David Kushner)	-0.7788	×
10	Something in the Orange	-1.0234	×

Interpretation: The score is a *relative* quantity. Its only purpose is to rank candidates for the same user. Absolute values are not comparable across users because $\hat{\theta}_u$ is defined up to a positive scalar.

Is θ Global or Per-User?

Answer

There is **no global** θ . Every user u has their own preference vector

$$\hat{\theta}_u = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{r}_u.$$

User	$\hat{\theta}_{u,1}$ (dance)	$\hat{\theta}_{u,2}$ (energy)	$\hat{\theta}_{u,3}$ (valence)
User A	0.2204	0.3260	0.3846
User B	0.0561	0.5852	0.3587
User C	$\vdots$	$\vdots$	$\vdots$

What is shared vs. per-user:

- **Shared:** $\mathbf{Z}$ (built from the catalog), $(\mathbf{Z}^\top \mathbf{Z})^{-1}$ (computed once).
- **Per-user:** $\mathbf{r}_u$ (affinity), $\hat{\theta}_u$ (preferences), score.

Model parameters: d per user, so $U \times d$ total.

Computation: invert once, then one $d \times d$ matrix-vector product per user.

Thank you. Questions?